

VERTEX ERP

ERP Software with an integrated workflow system

Built for Shahjalal Islami Bank PLC
Common Services Division

DEMONSTRATION · SEPTEMBER 2026

U Fintech · Harvest Partners · Nascenia

THE PROBLEM

Buying a laptop takes six weeks. Nobody can tell you why.

Not because anyone in Common Services is slow. Because the process has no system underneath it. A request is printed, signed, and physically carried between desks. It waits in somebody's tray while they are on leave. It is re-typed into a register, photocopied for the file, and chased by telephone.

By the time the goods arrive, four people have handled the same piece of paper and none of them can tell you where it has been.

What that costs you, every week

1 Nobody knows where anything is

Finding the status of one requisition means telephoning three people and hoping one of them remembers. Multiply that by every open request in the Division.

2 The Bank buys what it already owns

There is no way to see the store from the request. Stock sits on a shelf in one branch while another branch raises a purchase for the same item.

3 Prices creep and nothing notices

Nobody remembers what was paid last time. A supplier who quotes fifteen percent higher this year is simply paid fifteen percent more.

4 Approval limits depend on memory

The delegation of authority is a document in a drawer. Whether it is followed on a busy Thursday afternoon depends entirely on who is in the room.

5 An inspection becomes a two-week job

When the regulator asks for six months of procurement records, people stop their work and stand at a photocopier. The answer is assembled by hand, every time.

None of this is a failure of your people. It is what happens when a process that moves money has no system underneath it.

WHO IS DELIVERING THIS

U Fintech • Harvest Partners • Nascenia JV

A single joint venture, bidding as one entity and delivering as one team. One contract, one project manager, one technical account lead, named from day one. You will never be asked which company to call.

One accountable party

The joint venture carries the obligation as a whole. There is no arrangement under which one member can point at another, and no question of the Bank managing three relationships. You hold us to the contract; we hold ourselves to each other.

Combined capability

Local commercial and contractual presence in Bangladesh, with the Bank as our first priority in market. Product engineering behind the workflow engine, the permission model, the controls and the audit chain you are about to see. Engineering capacity and long-term local support, so the Bank is never dependent on a single office or a single time zone.

Through the delivery and beyond it

The same named team runs requirements, build, integration, migration, testing, training and go-live — and stays through the twelve-month warranty and any subsequent AMC term. The people in the room at kick-off are the people you call in month eleven.

Bidder qualification, experience certificates and corporate documents are as submitted with the technical bid.

TODAY

What we will show you

Most of this meeting is the working system, not this document. We would rather show you than describe.

Where we started

Who is delivering this

Live demonstration — one purchase, start to finish

Coverage against Annexure-A and Annexure-B, line by line

What we built beyond the specification

Technical specification — stack, architecture, hosting and disaster recovery

Delivery after the work order

Your questions

For the IT Division

A full technical section follows — the stack layer by layer, the three-tier architecture, hosting, and how disaster recovery replication works. Pages 12 to 15. We will reach it, so please hold technical questions until then if you would rather hear the answer than ask for it.

One thing worth saying before we start

The laptop this runs on is not connected to anything. No internet, no network, no cloud service. Everything you are about to see runs on this machine, from a database on this disk.

LIVE

One purchase, start to finish

Five people. A requisition, a tender, three bidders, a work order, a delivery and a payment.

On this laptop, with the wireless switched off.

Why this is not another system

It never touches the internet

Everything you just saw ran on this laptop with the wireless switched off. Installed at the Bank it works the same way — on your servers, on your own network, behind your own firewall. Your procurement data never leaves the Bank, never leaves the country, and never depends on a service somebody else operates.

The rules are yours, and you change them yourself

Approval limits, routing, escalation windows, who signs what — all configured by your own administrators on screen. When your delegation of authority changes, your system changes the same day. No change request. No waiting for us. No invoice.

It says no, so your people do not have to

Nobody has to refuse a colleague who asks them to approve their own request, or to release a price early, or to order more than was signed off. The system simply will not do it. That protects your staff as much as it protects your money.

Nothing can be quietly rewritten

Every record is cryptographically locked to the one before it. Any edit or deletion is detectable in seconds, and the system names the record that fails — for administrators too, which is the only version of that promise worth making.

It knows what kind of bank you are

Your Shariah Supervisory Committee is built in as an approval authority in its own right, with contract structures and screening rules the Committee owns. No general-purpose system does this.

It watches your money when nobody is looking

Seven tests run continuously over your own procurement, catching the patterns internal audit hunts for by hand — split purchases, price drift, supplier concentration, tenders with no real competition.

It is fast enough to actually be used

Every screen renders in well under a tenth of a second. A system people find slow is one they work around — and a control people work around is not a control.

You own it outright

Full documentation, database design, field mapping and API specifications handed to your IT Division. If you ever replace us, you keep everything.

All 25 modules, mapped line by line

We went back through your own Annexure-B and mapped every single sub-feature in it against what exists today.

25

of 25 modules
addressed

298

sub-features mapped
to your document

222

with a working screen
behind them today

74%

of the written scope,
built and testable

Running end to end — you can do it now  **133**

On screen — real data, action wired at
implementation  **89**

Scheduled for the build phase  **64**

Declared in our bid as needing
customisation  **12**

The full mapping is available

All 298 lines, module by module, showing exactly what runs today and what is scheduled. We can leave a copy with you.

You did not ask for this. We built it anyway.

Across your RFQ, Annexure-A and Annexure-B, the word “Shariah” does not appear once.

Your specification was written by the IT Division as a general procurement specification, and as a general procurement specification it is a good one. But Shahjalal Islami Bank PLC is a Shariah-based bank, and procurement at a Shariah-based bank is subject to its Shariah Supervisory Committee like everything else the Bank does.

So we built the Committee into the system. Not because it was asked for — because we do not believe we can hand an Islami bank a procurement platform that does not know what kind of bank it is.

The position we take, and will not move from

This system contains no Shariah rulings and makes none.

It holds the rules your Committee issues, applies them mechanically, and refers what they catch back to your Committee for a human decision. No other role can record a decision on the Committee’s behalf — not even the system administrator.

Working today, in front of you

The Committee as an approval authority · screening of vendors, contracts and work orders · Islamic contract structures recorded per agreement · halal certification tracked against canteen supply · a permanent decision record in the audit trail.

Your specification asks us to record procurement.

It never asks us to examine it. So we added seven tests that do.

These are the checks an internal audit function would run by hand, over a spreadsheet, perhaps twice a year. We run them continuously, over procurement you already record. No new data capture. No extra work for anyone. No integration.

Split purchase

Several small orders that together cross an approval limit

Price drift

Paying materially more than this Bank has paid before

Concentration

How much of your spend sits with a single supplier

Bid clustering

Bids arriving suspiciously close together in price

Repeat winner

One supplier winning an unusual share of tenders

Rapid approval

Approved too quickly for anyone to have read it

Single bid

An open tender that drew no competition at all

Running against this demonstration right now

7 questions raised · **1,22,53,546** under question

Every finding is a question, not an accusation. Each has an innocent explanation available and the wording says so. The system puts the question in front of a person with the evidence attached. It does not decide.

OUR COMMITMENT

More than quoted, at the price quoted

Everything in Annexure-A and Annexure-B is fixed scope at the price already submitted. Everything below is in addition, at no change to that price.

Already built, and neither was requested

Shariah governance · Procurement integrity

And any of these, sequenced into the delivery roadmap

Artificial intelligence, built in

An assistant over your own policy and live data. Ask in plain language, get the answer with the clause it came from. If you want it, it is included.

Supplier screening

Run your vendors through the same sanctions, PEP and CIB checks you already run on customers.

Bangladesh Bank inspection pack

One button: every procurement decision in a period, with approval chains, ready for inspection.

Read-only auditor access

Internal and external audit see everything, change nothing, and stop asking IT for exports.

Daily integrity seal

A dated code published each day, so you can prove to a regulator that records were never rewritten.

Vendor performance scoring

Delivery punctuality, rejection rates and penalty history, scored automatically from what you capture.

Budget commitment view

Committed but not yet invoiced — the number Finance needs and almost never has.

Mobile approvals

An approver clearing a requisition from a car, turning a two-day cycle into two hours.

Bangla interface

Full Bangla alongside English, switchable per user, for branch staff.

Delegation of authority screen

The DoP matrix as a screen you edit, with leave delegation and automatic escalation.

Contract drafting from templates

Contracts generated from approved templates rather than retyped.

Shariah audit reporting

A period report for your Supervisory Committee on everything it ruled on.

Tell us which matters most and we will sequence it first.

Sixty to ninety working days

You see a running system in week three, and again every fortnight after that. If something is going wrong you will know in week four, not week twelve.

- 1 Kick-off Week 1**
Everyone in one room. We ask for one thing: the names of the people who own each decision.
- 2 Your rules, configured with you Weeks 1-3**
Your procurement policy and delegation matrix, entered in the builder you just saw, and signed off before anything is locked.
- 3 Build and configure Weeks 3-8**
The demonstration becomes your system. It is the foundation, not something we throw away.
- 4 Integration Weeks 6-10**
CBS, HRMS, DMS, e-mail, SMS and IAM through the middleware tier, each tested alone first.
- 5 Your data moves in Weeks 8-11**
Vendors, contracts, assets, stock and budgets. You hand us your lists — nobody retypes years of records.
- 6 You test it Weeks 10-12**
A separate environment mirroring production, with acceptance criteria agreed in advance.
- 7 Training and go-live Weeks 12-13**
Three streams: IT Division, Common Services, branch and head office. Then live, division by division.

Then twelve months of warranty

24x7 helpdesk, SLA-backed response tiers, and any change a regulatory shift requires at no cost.

What it is built on

Every element chosen against three tests: proven in regulated financial deployments, free of proprietary licensing, and maintainable by any competent team — not only by us.

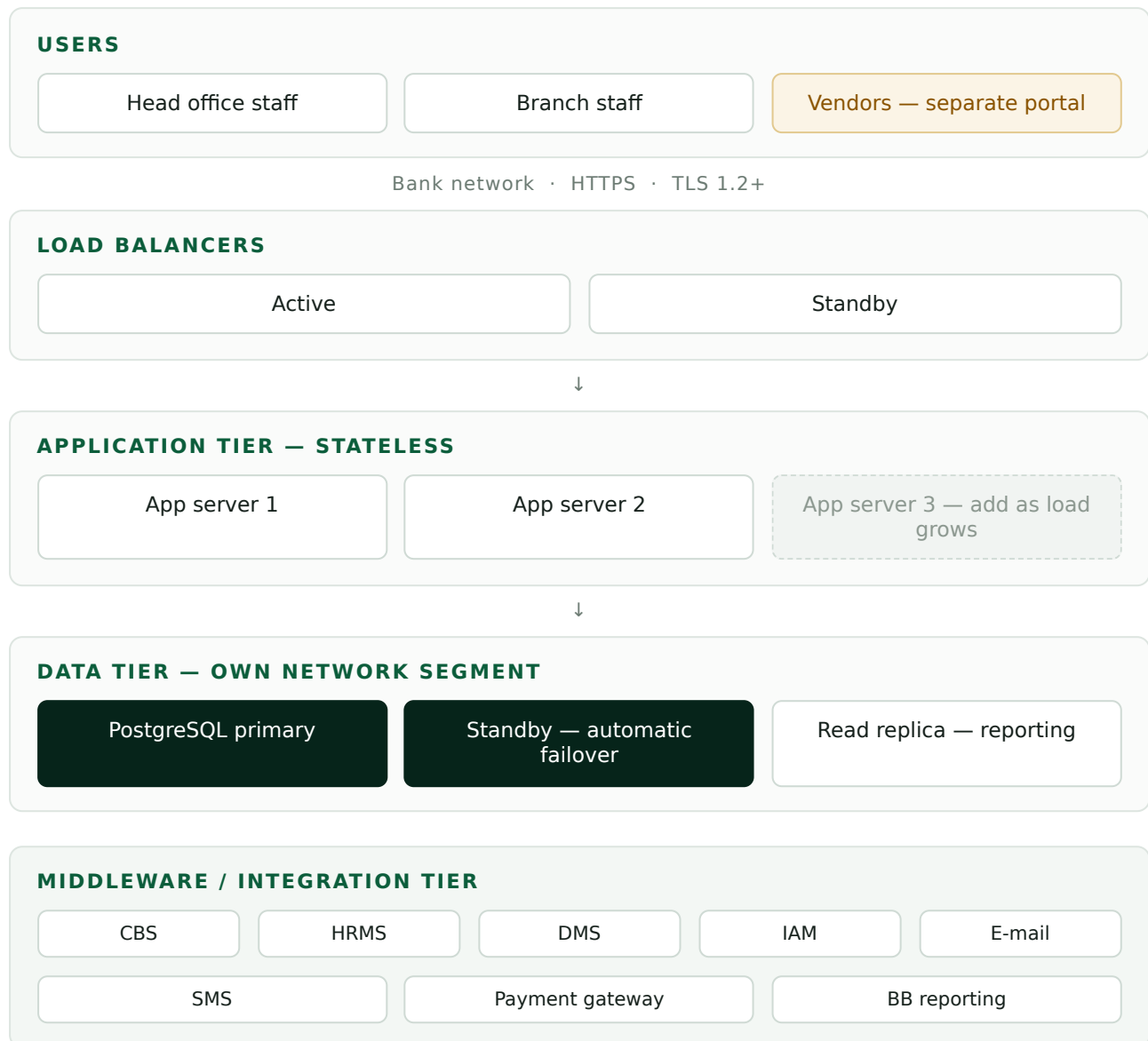
Layer	Technology	Why this one
Operating system	Linux — Ubuntu Server LTS 22.04+	Five-year support per release, hardened, no per-server licence
Runtime	Node.js 22 LTS	Long-term support, single-threaded event model suited to I/O-bound banking workloads
Language	TypeScript 5, strict mode	Static typing catches an entire class of bug at build time — it matters when the code moves money
Application	Node.js service tier, containerised	Stateless, so instances can be added or removed without downtime
Interface	React 19, server-rendered	Screens render on the server; the browser receives finished HTML, so branch machines stay light
Database	PostgreSQL 15+	ACID, row-level security, streaming replication and point-in-time recovery, all native
Data access	Prisma ORM, typed	Every query type-checked against the schema at build time; no untyped string SQL
API	REST with webhook support	REST for synchronous calls to CBS; webhooks so your systems are told of events rather than polling
Authentication	Signed session cookie, script hashing	Integrates with your central IAM for single sign-on where you want it
Integrity	SHA-256 hash chain	Each audit record carries the fingerprint of the one before it
Packaging	Docker containers	The same image runs in UAT and production — no “it worked in staging”
Testing	Automated browser and control suites	52 control assertions and 99 screen checks before every release

No proprietary licensing anywhere in this stack

No per-seat fees, no per-core fees, no commercial database or application-server licence. Nothing here bills the Bank again as it grows.

Three tiers, inside your building

Presentation, business logic and data are separated, with a dedicated middleware tier owning every connection to the rest of the Bank's estate.



Why the middleware tier exists

One layer owns every external connection. When your CBS upgrades its API, one place changes and procurement logic is never touched.

Why the application tier is stateless

No session is held in server memory, so any server can answer any request — which is what makes horizontal scaling and zero-downtime deployment possible.

Where this actually runs

On your infrastructure, in your data centre, under your control. No component of this system is hosted by us, and no data leaves the Bank.

Production — primary data centre

2 application servers (16 vCPU / 64 GB, load balanced) · 2 database servers (32 vCPU / 128 GB, active-passive) · 1 middleware server (8 vCPU / 32 GB) · 4 TB usable, RAID 10

Streaming replication — frequency set by the Bank

Disaster recovery site

1 application + 1 database server, matching specification, 4 TB replicated

Asynchronous replication

Far disaster recovery site

1 application + 1 database server, 4 TB archival, sized for five-year retention

Deployment is your choice, and reversible

On-premise, private cloud or public cloud — the same container image runs in all three. Start on-premise and move later without the application being rewritten.

What we host: nothing

No procurement data is transmitted outside the Bank's network, and none leaves Bangladesh. There is no vendor cloud service that can fail, be suspended, or change its terms.

The demonstration you are watching

Today the application, the database and the browser all run on this one laptop with the wireless switched off, to show that no external dependency exists. Installed at the Bank it is the same software on the servers above, reached over your own network like any other internal system.

Backup, archival and recovery

Administrator-controlled backup from inside the system, on your schedule, with manual backup on demand — deliberately separate from infrastructure backup so there are two independent layers. Archival retention is configured against Bangladesh Bank requirements.

How the data reaches the other sites

Continuously, over your own encrypted link, with the Bank setting how far behind the copies are allowed to fall.

Production — primary database

Every change is written to a transaction log **before** it is applied. That log is the thing that travels.

↓ Streaming replication over a dedicated link or site-to-site VPN · encrypted in transit (TLS 1.2+)

Disaster recovery — hot standby

Receives the log continuously and replays it, staying seconds behind production. Promoted to primary on failure.

↓ Asynchronous replication · same encrypted path

Far disaster recovery — archival

A third copy at geographic distance, for the case where a whole region is lost.

What actually travels

Not files, and not a nightly dump. The database writes every change to a transaction log before applying it, and that log is streamed to the standby as it is produced. The standby replays it, so it holds a continuously current copy rather than last night's.

You set how much you are willing to lose

Run the DR link **synchronously** and a transaction is not confirmed until the standby has it — zero data loss, at the cost of a few milliseconds per write. Run it **asynchronously** and writes confirm immediately, with the standby a few seconds behind. This is your recovery point objective, it is a setting, and Annexure-A commits to your IT Division owning it.

Nothing crosses a public network

Replication runs over the Bank's own dedicated link or site-to-site VPN, encrypted end to end. No replication traffic touches the internet and no copy of Bank data leaves the country.

Recovery, and proving it works

On failure the standby is promoted and the application tier reconnects — no reinstallation, no restore from tape. Alongside this, point-in-time recovery lets the Bank rewind the database to any moment within its retention window, which is what you want when the fault is a bad change rather than a dead machine. We recommend and support a live failover test at least annually, because DR that has never been tested is an assumption.

We are not the lowest bid. Here is what you get instead.

1 **The system refuses. Not the policy.**

In most systems a control is a rule in a manual that people are trusted to follow. In this one the computer simply will not do it. That is the difference between a policy and a guarantee.

2 **We read your business, not just your specification**

Nobody asked us for Shariah governance or procurement integrity. We built both, before this meeting, because you are an Islami bank that buys things.

3 **We are already doing this for financial institutions in Hong Kong**

This is not our first implementation and not a product we are building on your money. It is running today, in one of the most heavily regulated financial markets in the world.

4 **We move faster than anyone else you will meet**

What you have just seen was built, tested and hardened in days. Ninety working days to go live is not an optimistic number for us. It is a comfortable one.

5 **You own it. You are never locked in.**

Your workflows, your reports, your roles, your rules — all configured by your own administrators. Full documentation, database design and API specifications handed over. If you ever replace us, you keep everything.

6 **More than quoted, at the price quoted**

Two modules already delivered beyond scope. Twelve more, including artificial intelligence, available at no change to the licence price.

Thank you

We would rather answer your questions than present to you. Ask us anything — including about the parts we have not built yet.

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